

TLYB'S Gigafactory — Operational Description

Reference plant: Tesla Gigafactory Nevada (GF1), at approximately half-scale

Document purpose: Operational ground truth for the HEPHAESTUS multi-agent AI system simulator

Author note: This is the foundational reference document. Every process, equipment specification, cycle time, and KPI below is grounded in published lithium-ion battery manufacturing literature, Tesla's publicly disclosed operations, and industry-standard practice. Numbers are illustrative within realistic ranges; they are the kind of figures you would defend in an interview.

1. EXECUTIVE OVERVIEW

What TLYB'S Is

TLYB'S Gigafactory is a vertically integrated lithium-ion battery and EV powertrain manufacturing facility. It produces three primary product families on co-located lines: lithium-ion cells (the foundational product), battery packs (cells integrated into vehicle-ready assemblies), and complementary powertrain components (motors, inverters, drive units — modeled as adjacent capacity for Phase 2+).

For the purposes of HEPHAESTUS Phase 1, the AI system focuses exclusively on the cell production line, with awareness of pack assembly as the immediate downstream consumer.

Scale and Capacity

TLYB'S is sized at approximately half the operational scale of Tesla Gigafactory Nevada in its 2024 configuration:

Dimension	Tesla GF1 (reference)	TLYB'S (modeled)
Annual cell capacity	~35 GWh	~18 GWh
Annual pack capacity	~500,000 vehicles	~250,000 vehicles
Direct workforce	~7,000	~3,500
Facility footprint	~5.3 million sq ft	~2.8 million sq ft
Operating mode	24/7, 3 shifts	24/7, 3 shifts
Cell chemistry	NCA (Panasonic)	NCM 811 (in-house)
Cell format	2170, 4680	4680 only

Why 4680 Cells

TLYB'S produces only 4680 format cells (46mm diameter × 80mm tall). This is the format Tesla, Panasonic, and LG have all converged on for next-generation EVs because of its energy density, manufacturing throughput per cell, and lower joining-point count at the pack level. Producing a single format simplifies the line and improves yield — a deliberate design choice for a new entrant.

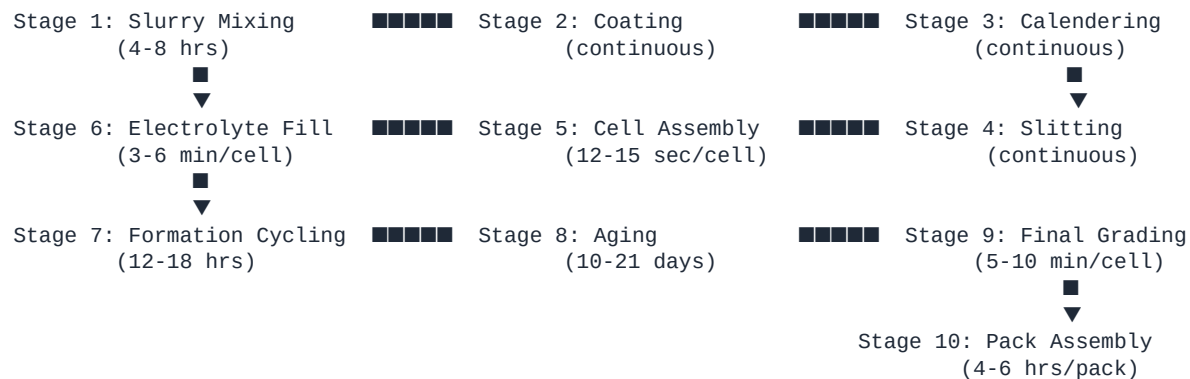
The AI System's Mandate

HEPHAESTUS is responsible for one primary objective at TLYB'S: **maximizing yield on the cell production line**. Every cell that completes Formation cycling and fails Final Grading represents approximately \$35-50 in wasted materials and 18-24 hours of facility time. At our scale, even a 1% yield improvement saves the company on the order of \$15-20 million annually.

The supporting agents (HERMES for procurement, THEMIS for compliance) exist in service of this objective: procurement quality drives input variance which drives yield, and compliance traceability ensures every yield decision is defensible.

2. PRODUCTION LINE OVERVIEW

The TLYB'S cell production line is a continuous-flow process with nine major stages, plus pack assembly as a tenth stage downstream. Total time from raw material to shipped cell is approximately 21 days; total active processing time per cell is approximately 24 hours, with the remainder being formation, aging, and quality stabilization.



Critical insight for HEPHAESTUS design: The point of maximum AI leverage is between Stage 4 (Slitting) and Stage 7 (Formation Cycling). After Stage 7, a cell has consumed ~18 hours of facility time and significant energy. Predicting failure before Stage 7 saves the most downstream cost. After Stage 8 (Aging, 10-21 days), the prediction is moot — the cell is already old.

3. STAGE-BY-STAGE OPERATIONAL DESCRIPTION

Stage 1: Slurry Mixing

What happens: Active cathode material (NCM 811 — Nickel-Cobalt-Manganese, 80% nickel) and active anode material (synthetic graphite + silicon oxide) are blended separately with binders (PVDF for cathode, CMC/SBR for anode), conductive additives (carbon black), and solvents (NMP for cathode, water for anode) in industrial planetary mixers. The output is a viscous slurry of precisely controlled rheology.

Equipment: 8 planetary vacuum mixers per line (4 cathode, 4 anode). Each mixer holds approximately 1,500 kg per batch. Mixers from PRIMIX or LARGER are standard.

Cycle time: 4-8 hours per batch depending on chemistry and viscosity targets.

Throughput: Each mixer produces approximately 4,500 kg of slurry per 24-hour period. The line consumes approximately 30,000 kg of slurry per day at full capacity.

Critical process parameters:

- Solids loading (mass percent active material in slurry): cathode 70-78%, anode 45-55%
- Viscosity: cathode 3,000-8,000 cP, anode 2,000-5,000 cP
- Particle size distribution (D50 of active material)
- Mixing energy (kWh/kg)
- Temperature during mixing (controlled to 22-25°C)

Quality metrics:

- Slurry homogeneity (measured by sampling and laser diffraction)
- Absence of agglomerates >100 µm
- Stability over hold time (slurry must remain pumpable for 8-12 hours)

Common failure modes:

- Binder settling (slurry separates during hold) → entire batch scrapped, ~\$15,000 loss per cathode batch
- Agglomeration (particles clump) → coating defects downstream
- Contamination (any metallic particle) → potential cell short circuit later, catastrophic safety risk

Material costs (illustrative):

- NCM 811 cathode powder: ~\$28/kg
- Synthetic graphite anode: ~\$12/kg
- Silicon oxide additive: ~\$45/kg
- PVDF binder: ~\$30/kg
- NMP solvent: ~\$3/kg (~95% recovered and recycled)

KPIs tracked:

- Batches mixed per shift
- Batch acceptance rate (target >98%)
- Solvent recovery rate (target >95%)

- Mixer downtime hours per month

Stage 2: Electrode Coating

What happens: Slurry is pumped through a precision slot-die coater onto a thin metal foil substrate — copper foil for the anode (8-10 μm thick), aluminum foil for the cathode (12-15 μm thick). The coated foil moves continuously through long drying ovens that evaporate the solvent, leaving a uniform layer of active material bonded to the foil.

Equipment: Slot-die coaters (M-TEK, Hirano Tecseed are leading suppliers) with web widths of 1.0-1.5 meters. Each line has 2 coaters (one cathode, one anode). Drying ovens are 30-50 meters long, divided into temperature zones.

Cycle time: Continuous process. Web speed is typically 30-80 meters per minute. A single coil of foil (~5,000 meters) takes 60-150 minutes to process.

Throughput: Each coater produces approximately 200,000 square meters of coated electrode per day at full capacity.

Critical process parameters:

- Coating thickness (after drying): cathode 60-90 μm , anode 70-100 μm
- Coating thickness uniformity (cross-web, target $\pm 1.5 \mu\text{m}$)
- Web tension (controlled to prevent stretching)
- Drying temperature profile (zone-by-zone, peak 130-150°C)
- Solvent retention in dried coating (target <0.5% by mass)

Quality metrics:

- Areal mass loading (mg/cm^2) — directly determines cell capacity
- Coating defects (pinholes, streaks, edge effects)
- Adhesion strength (peel test, target >50 N/m)
- Surface roughness

Common failure modes:

- Coating thickness drift (slot-die gap changes) → entire coil out of spec
- Solvent flash-off too rapid (cracks in coating) → batch failure
- Foil tension variation (wrinkles, edge curl)
- Foreign particles embedded in coating (potential shorts in finished cell)

Why this stage matters for HEPHAESTUS: Coating thickness uniformity is the single strongest predictor of cell capacity variance in the finished product. A 5% coating thickness variation can produce a 5% capacity variation, which means cells from the same batch grade out differently in Stage 9. This is your model's most predictive input feature.

KPIs tracked:

- Square meters coated per shift
- First-pass coating yield (target >95%)
- Coating thickness CpK (process capability)
- Average defect density (defects per square meter)

Stage 3: Calendering

What happens: Coated electrode passes through heated steel rollers under high pressure (4,000-8,000 N/cm). This compresses the porous coating to a precise final density and thickness, improving electrical contact between active particles and increasing volumetric energy density.

Equipment: Calendering machines (Hitachi, Sumitomo Heavy Industries). Heated rollers, typically 60-80°C operating temperature. Force is hydraulically controlled.

Cycle time: Continuous. Web speed matched to coating speed.

Critical process parameters:

- Roller force (the most sensitive parameter — small changes have large effects)
- Roller temperature
- Web tension before and after rollers
- Final electrode density (target 3.4-3.7 g/cm³ for cathode, 1.6-1.8 g/cm³ for anode)
- Final electrode thickness (after compression)

Quality metrics:

- Density uniformity across web
- Absence of cracking (over-calendered)
- Absence of springback (under-calendered)
- Electrode porosity (target 25-35%) — too low restricts ion transport, too high hurts energy density

Common failure modes:

- Roller wear (asymmetric calendering) → batch out of spec, expensive roller replacement
- Force drift (hydraulic system) → gradual density change, missed by single-point QC
- Foil cracking (over-pressure) → catastrophic, scrap entire coil

Why this stage matters: Calendering density directly affects cell impedance, which affects formation behavior, which affects yield. The relationship is nonlinear and supplier-dependent (different binders calender differently).

KPIs tracked:

- Calendering force CpK
- Roller condition (measured weekly via profilometry)
- Density distribution per coil

Stage 4: Slitting

What happens: The wide calendered electrode coil is cut longitudinally into multiple narrower strips, each sized to the specific width required for the 4680 cell format.

Equipment: Slitter-rewinder machines with precision rotary or razor blades. Blade-on-bottom-anvil cutting is standard for clean edges.

Cycle time: Continuous. Web speed typically 80-150 m/min through slitter (faster than coating because no drying required).

Critical process parameters:

- Blade sharpness (degrades over usage — typical blade life is 50,000-100,000 meters of cut)
- Cut accuracy (target ± 0.1 mm)
- Burr height on cut edge (target < 5 μm — burrs cause shorts)
- Edge tape application (some designs require edge insulation tape applied here)

Quality metrics:

- Strip width consistency
- Edge burr profile
- Absence of foreign material in cut

Common failure modes:

- Blade degradation creates burrs → cells produced from this material will have higher short-circuit failure rates in Formation
- Mis-tension causes lateral wander → some strips out of width spec, scrap

KPIs tracked:

- Slit yield (target $> 98\%$ — slit losses include trim and rejects)
- Blade change frequency
- Edge quality grade per coil

Stage 5: Cell Assembly

What happens: This is where individual cells take physical form. Strips of cathode, separator, and anode are wound into the cylindrical "jelly roll" structure that defines a 4680 cell. The jelly roll is then inserted into a steel can, the bottom is welded shut, and a current collector is laser-welded to the cell tab.

Equipment: High-speed winding machines (Manz AG, KOEM, CIS), can welders, laser welders. This is the most capital-intensive stage of the line — a single winding line costs \$15-25 million.

Cycle time: 12-15 seconds per cell. This is the line's pacing stage.

Throughput: Each winding machine produces ~5,000 cells per day. TLYB'S has 24 winding machines, for a theoretical maximum of 120,000 cells per day. With realistic uptime and yield, sustainable production is ~95,000 cells per day.

Critical process parameters:

- Winding tension (must be precisely controlled — too loose causes voids, too tight causes electrode damage)
- Winding speed
- Separator alignment (must extend slightly beyond electrodes on both ends)
- Weld penetration depth and uniformity
- Can insertion force

Quality metrics:

- Jelly roll geometry (concentricity, diameter)
- Weld strength (sample tested)
- Internal resistance pre-electrolyte (initial measurement, predicts cell health)
- Foreign material detection (X-ray inspection of every cell)

Common failure modes:

- Winding tension excursion → jelly roll deformation, cell fails Formation
- Weld defects → cell fails leak test or develops high impedance over time
- Foreign particles introduced during assembly → potential short circuit (CATASTROPHIC — can cause thermal runaway)

Why this stage matters: Every cell from this point forward has a unique serial number and is individually tracked. The yield prediction starts to become very accurate here because individual cell measurements (resistance, dimensions, weight) are available.

Safety note: This stage is the highest contamination-risk point in the entire process. Cleanroom protocols (ISO Class 7 or better) are mandatory. A single 50 µm metal particle inside a cell can cause thermal runaway months later in a customer's vehicle.

KPIs tracked:

- Cells assembled per shift per machine
- Assembly first-pass yield (target >99%)
- Weld inspection acceptance rate
- Foreign material rejection rate

Stage 6: Electrolyte Fill

What happens: Each cell is filled with a precise volume of liquid electrolyte (typically LiPF₆ salt dissolved in a mixture of organic carbonates — EC, DMC, EMC). The electrolyte must wet the entire jelly roll, which requires controlled vacuum filling to displace air. The cell is then sealed.

Equipment: Vacuum filling machines (Manz, Hitachi). Each station processes one cell at a time.

Cycle time: 3-6 minutes per cell. This stage requires multiple cells in parallel to match upstream throughput.

Throughput: TLYB'S runs approximately 400 fill stations in parallel to match winding output.

Critical process parameters:

- Electrolyte volume (target $\pm 2\%$ of nominal)
- Vacuum level during fill
- Fill rate (too fast causes incomplete wetting)
- Soak time after fill (allows electrolyte to penetrate jelly roll)
- Seal integrity (laser weld of fill port)

Quality metrics:

- Cell weight after fill (validates electrolyte volume)
- Leak rate (helium leak detection on sampling basis)
- Electrolyte distribution (inferred from impedance after soak)

Common failure modes:

- Underfill → cell fails Formation
- Overfill → seal failure under pressure during cycling
- Incomplete wetting → high impedance, capacity loss
- Seal defects → electrolyte leakage, cell scrap

Material cost note: Electrolyte is approximately \$8-12 per 4680 cell — a meaningful portion of total cell cost.

KPIs tracked:

- Fill accuracy CpK
- Seal acceptance rate
- Cells filled per shift

Stage 7: Formation Cycling

What happens: This is where a freshly assembled cell becomes a working battery. Each cell is connected to a programmable charger and slowly charged for the first time. During this initial charge, the SEI (Solid Electrolyte Interphase) layer forms on the anode — a microscopic protective film that determines almost everything about the cell's future behavior: cycle life, calendar life, safety profile, capacity stability.

The cell is then partially discharged, gas formed during SEI creation is vented (in some designs), and the cell goes through several additional precisely controlled cycles to stabilize the SEI.

Equipment: Massive formation racks — each cell occupies a single slot in a rack. TLYB'S has approximately 25,000 formation slots running simultaneously. Each slot has its own programmable channel (voltage, current

control) and temperature monitoring. Equipment suppliers: Pec, Maccor, Arbin.

Cycle time: 12-18 hours per cell. THIS IS THE BOTTLENECK OF THE ENTIRE LINE.

Throughput: ~95,000 cells per day complete Formation at full capacity. This matches the upstream Assembly throughput by design.

Critical process parameters:

- Formation current (typically C/20 to C/10 — very slow charge)
- Formation voltage limits
- Temperature during formation (controlled to ~25-30°C)
- Cycle profile (multi-step charge/discharge sequence)
- Rest periods between steps

Quality metrics:

- Coulombic efficiency on first cycle (target >90%)
- Capacity at end of Formation (target 95-105% of nominal)
- Internal resistance after Formation
- Voltage relaxation behavior (predicts cycle life)

Common failure modes:

- Poor SEI formation → low cycle life, cell will fade quickly in service
- Internal short circuit develops during first charge → cell scrap, sometimes thermal event
- Electrolyte decomposition outside expected range → gas formation, swelling, scrap

Why this stage matters massively for HEPHAESTUS:

This is the single most expensive stage in terms of facility time. Every cell that enters Formation consumes 12-18 hours of a \$5,000+ rack slot. If a cell is going to fail Final Grading, predicting it BEFORE it enters Formation saves the most cost.

The yield prediction model's primary job: Given measurements from Stages 2-6 (coating thickness uniformity, calendaring density, slitting edge quality, assembly weld inspection, pre-fill internal resistance, post-fill weight), predict whether this specific cell will pass or fail Final Grading 19 hours from now.

Cost of getting it wrong:

- False positive (scrap a good cell): ~\$45 wasted material
- False negative (let a bad cell through): ~\$65 wasted material + 14 hours of rack time + downstream contamination of statistics

Energy cost: Formation cycling consumes enormous amounts of electricity. TLYB'S formation operations consume ~80 MW continuously. This is also where energy management AI (Phase 3) becomes valuable.

KPIs tracked:

- Formation throughput (cells per hour)

- Formation rack utilization (target >85%)
- Cells failing Formation as percentage (target <3%)
- Average energy per cell (kWh)

Stage 8: Aging

What happens: Cells are stored at controlled temperature (typically 25-45°C) for 10-21 days. During this time, the SEI layer continues to stabilize, residual stresses in the jelly roll relax, and any latent defects manifest. Cells are periodically checked for voltage drift (self-discharge) — abnormally high self-discharge indicates an internal defect.

Equipment: Large climate-controlled warehouses with cell trays. Voltage monitoring infrastructure on a sampling basis.

Cycle time: 10-21 days per cell.

Critical process parameters:

- Storage temperature
- Storage state of charge (typically ~30%)
- Total aging time

Quality metrics:

- Self-discharge rate (mV/day) — primary aging metric
- Voltage stability
- Capacity retention vs. pre-aging measurement

Common failure modes:

- Latent internal short circuits develop → cell self-discharges rapidly, scrap
- Capacity fade beyond spec → downgrade or scrap
- Swelling (gas formation) → safety issue, scrap

Why this stage matters for HEPHAESTUS: The aging stage is where the model's predictions are validated. A cell predicted to fail at Stage 7 that survives Aging is a false positive. A cell predicted to pass that fails Aging is a false negative. This continuous validation is what THEMIS tracks for model accuracy drift.

KPIs tracked:

- Cells in aging inventory
- Aging-stage failure rate (target <2%)
- Average aging duration (vs. minimum required)

Stage 9: Final Grading

What happens: Every cell undergoes complete electrical characterization: capacity test, internal resistance measurement, voltage profile during discharge, self-discharge confirmation. Based on results, each cell is binned into a grade (A, B, C) and matched with cells of similar characteristics for pack assembly.

Equipment: Automated test cabinets (Pec, Maccor). Each cell tests for 2-4 hours.

Cycle time: 5-10 minutes per cell of active testing per characteristic, but cells are tested in parallel.

Throughput: Matches upstream — ~95,000 cells per day.

Critical metrics measured:

- Rated capacity (Ah)
- Energy capacity (Wh)
- Internal resistance ($m\Omega$) at 1 kHz AC
- DC resistance at multiple discharge rates
- Capacity retention vs. theoretical
- Self-discharge rate (final confirmation)

Quality grades:

- **Grade A** (premium): top 60-70% of cells, used in performance vehicles or premium configurations
- **Grade B** (standard): middle 20-30%, used in standard vehicle packs
- **Grade C** (commercial): bottom 5-15%, used in stationary storage applications
- **Scrap:** outside acceptable bounds — recycled

KPIs tracked:

- Overall first-pass yield (Stage 9 result is the headline number)
- Grade A percentage
- Average capacity vs. nominal
- Scrap rate at Final Grading

Industry benchmark: Tier-1 manufacturers achieve 92-96% Final Grading yield (i.e., 4-8% of cells that reach Final Grading are scrapped). Improving this by even 1% is worth \$15-20M annually at TLYB'S scale.

Stage 10: Pack Assembly (Downstream Consumer)

What happens: Cells are matched in groups based on grading parameters (capacity, resistance, voltage) and assembled into modules, then packs. The 4680 format is designed for structural integration into the vehicle, so the pack itself becomes a chassis component.

Equipment: Robotic pick-and-place systems for cell positioning, structural adhesive dispensers, wire-bond machines for inter-cell connections, automated pack-testing stations.

Cycle time: 4-6 hours per pack (including cure time for structural adhesive).

Throughput: TLYB'S can assemble ~700 packs per day.

Cells per pack: Approximately 600-1,000 cells per pack depending on vehicle configuration.

Critical process parameters:

- Cell matching tolerance (cells within a pack should be within $\pm 2\%$ of each other on key metrics)
- Adhesive bond quality
- Inter-cell connection resistance
- Pack-level insulation resistance

Why this matters for the upstream line: Pack yield is gated by cell matching. If the cell line produces wide variance, the pack line has to scrap or downgrade cells that don't match well. Tight upstream process control → tight cell-to-cell variance → high pack yield → less scrap. The connection runs all the way from Slurry Mixing.

KPIs tracked:

- Packs assembled per shift
- Pack first-pass test yield
- Cells consumed per pack (vs. theoretical minimum)
- Cell-to-cell matching CpK within packs

4. SUPPORTING OPERATIONS

Raw Material Procurement (HERMES domain)

Primary materials sourced:

Material	Annual volume	Typical lead time	Geopolitical exposure
Lithium hydroxide	~3,500 tonnes	90-180 days	Australia 50%, Chile 30%, China 20%
Nickel sulfate	~5,500 tonnes	60-120 days	Indonesia 60%, Russia 15%, others 25%
Cobalt sulfate	~700 tonnes	90-150 days	DRC 70% — ESG/conflict mineral concern
Manganese sulfate	~1,200 tonnes	30-60 days	South Africa 40%, China 30%
Synthetic graphite	~7,000 tonnes	60-90 days	China 75% — export risk
Copper foil	~1,800 tonnes	30-60 days	Multi-source, lower risk
Aluminum foil	~1,500 tonnes	30-45 days	Multi-source
Electrolyte (premixed)	~12 million L	30-90 days	Limited specialty suppliers
Separator film	~25 million m ²	60-90 days	Japan 60%, Korea 25%

Supplier quality scoring dimensions:

1. Price stability
2. Lead time consistency
3. Lot-to-lot quality variance (critical for yield)
4. ESG profile (especially cobalt — must demonstrate conflict-mineral-free)
5. Geographic concentration risk
6. Financial stability

Quality Control (cross-cutting)

QC operates at three layers:

1. **In-line QC** — automated measurement at every process stage, real-time
2. **Statistical QC** — sampling-based deeper inspection on a defined frequency
3. **Lot QC** — destructive testing on a small sample per lot

THEMIS maintains the audit trail across all three layers, ensuring every cell shipped can be traced back through every process step, every operator, every material lot, every measurement.

Regulatory Compliance (THEMIS domain)

Applicable frameworks:

- **UN 38.3** — battery transportation safety (cells must pass altitude, thermal, vibration, shock, external short circuit, impact/crush, overcharge, forced discharge tests)
- **IATF 16949** — automotive quality management system (mandatory for any auto OEM customer)
- **ISO 14001** — environmental management
- **ISO 45001** — occupational health and safety
- **OSHA standards** — workplace safety (US operations)
- **Dodd-Frank Section 1502** — conflict minerals reporting (cobalt, tantalum, tungsten, tin from DRC region)
- **EU Battery Regulation 2023/1542** — carbon footprint disclosure, due diligence, recycled content (becomes mandatory for EU sales)
- **California Proposition 65** — chemical exposure labeling

Every regulatory framework imposes audit trail requirements. THEMIS exists primarily to maintain this trail in a way that survives external audit.

Maintenance Operations

Maintenance strategy at TLYB'S:

- **Reactive** for low-criticality equipment

- **Time-based preventive** for most major equipment (calendared schedules)
- **Condition-based** for critical bottleneck equipment (formation racks, coaters, winders) — sensor-driven
- **Predictive (AI-driven)** — the long-term opportunity for HEPHAESTUS Phase 3+

Maintenance windows: scheduled during planned downtime. Production runs 24/7 with one 8-hour planned maintenance window per week per line.

5. PLANT-LEVEL KEY PERFORMANCE INDICATORS

These are the headline metrics tracked at the plant level, reported daily to operations leadership and continuously to HEPHAESTUS:

KPI	Target	Industry benchmark	TLYB'S current (simulated)
Overall cell yield (Final Grading pass rate)	>94%	92-96%	93.2%
Daily cell production	95,000	—	91,400 average
Daily pack production	700	—	685 average
OEE (Overall Equipment Effectiveness)	>75%	65-75%	71%
Energy per cell (kWh)	<0.45	0.4-0.5	0.43
Cost per cell (\$)	<\$48	\$45-55	\$49.20
Scrap value as % of materials	<3.5%	3-5%	3.8%
On-time delivery to pack line	>98%	—	97.1%
Safety incidents per million hours	0 (target)	1.2 industry avg	0.4
Compliance audit findings (per quarter)	0 critical	—	0 critical, 2 minor

6. WHERE HEPHAESTUS LIVES IN ALL OF THIS

Now that the operations are described, the AI system's role becomes specific and grounded.

FORGE — Production Intelligence Agent

- Monitors real-time telemetry from Stages 1-6 (Mixing through Electrolyte Fill)
- Runs yield prediction on every cell as it exits Stage 6, BEFORE entering Stage 7 (the 18-hour Formation bottleneck)
- For cells with >70% predicted failure probability, recommends early scrap to operations

- Tracks cumulative yield trends, alerts on drift
- During Formation, monitors voltage curves for early warning signs

HERMES — Procurement Intelligence Agent

- Continuously scores incoming material lots on quality variance metrics
- Maintains supplier scorecards across price, quality, lead time, ESG, geographic risk
- When a supplier's quality variance trends upward, raises an early procurement-shift recommendation
- Tracks long-lead-time materials (lithium, nickel, cobalt) and recommends strategic reorder timing

THEMIS — Compliance & Intelligence Agent

- Maintains traceability for every cell from raw material lot to final disposition
- Logs every FORGE prediction and its actual outcome
- Calculates model performance metrics (precision, recall, drift over time)
- Flags when the yield prediction model needs retraining
- Generates compliance-ready reports for UN 38.3, IATF 16949, and conflict minerals requirements
- Powers the natural-language interface that operations leadership uses to query the system

7. WHAT'S OUT OF SCOPE FOR PHASE 1

For clarity, the following are real parts of a gigafactory but are NOT modeled in the HEPHAESTUS Phase 1 simulator:

- Motor and inverter production lines (Phase 2+)
- Full pack assembly simulation (modeled only as downstream consumer of cells)
- Workforce scheduling and labor planning
- Detailed energy management and grid optimization (Phase 3+)
- Recycling and second-life cell operations
- Multi-site coordination (TLYB'S is single-site for now)
- Customer order management and shipping logistics

Each of these is a legitimate Phase 2/3/4 extension of HEPHAESTUS, with clear architectural hooks already built in.

8. DOCUMENT VERSION

- Version 1.0 — Initial operational reference for HEPHAESTUS v2 design
- Date: May 2026
- Author: HEPHAESTUS architecture working session

- Reference plant: Tesla Gigafactory Nevada (publicly disclosed operations as of 2024-2025)
- Next revision: After Phase 1 simulator is built and validated against this reference

This document defines the operational reality the HEPHAESTUS AI system is responsible for. Every simulator decision, every agent behavior, every dashboard metric should be traceable back to a specific section above. If something in the simulator doesn't correspond to something here, either the simulator is wrong or this document needs updating.